Community Collage — Installation & Operator Guide

Purpose: Run the physical four-choice ball-voting exhibit and its full-screen visualisation on a different Windows computer.

Current status (28 October 2026): The Arduino-to-website voting connection has been tested and works. Both distance sensors initialise and the hand sensor detects a hand, but the servo **has not yet been confirmed to move reliably with all components connected**. See **Power and servo troubleshooting** before relying on automatic ball release at an event.

1. What to bring

Hardware

- Arduino Uno-compatible board (DFRduino Uno v3.0), with its USB data cable
- Two VL53L0X time-of-flight distance sensor modules with accessible **XSHUT** pins
- SG90 9 g micro servo and the ball-release mechanism
- Breadboard and jumper wires
- Physical voting apparatus and balls
- **Recommended:** a separate regulated **5 V, 1–2 A** supply for the servo, plus suitable connectors, with common ground to the Arduino
- Windows laptop with USB port, charger, Chrome or Edge, and permission to install software
- Optional: USB hub/adapter if the laptop lacks the correct port

Files

Keep these together in one folder (for example `Desktop\Community-Collage`):

```
Community-Collage/
  README.md
  community-collage-arduino-no-popups.html
  CommunityCollage_Combined/CommunityCollage_Combined.ino
  https_emaie3_github_io_2850_ (1).png   [optional QR image; not included unless supplied]
```

Important: The HTML refers to a QR-code image called `https_emaie3_github_io_2850_ (1).png`. Copy that exact image into the same folder as the HTML if the QR code must appear. The exhibit otherwise runs without the image. The webpage also loads Google Fonts from the internet; if offline, browser fallback fonts will appear.

2. Software to install on a NEW Windows laptop

1. **Google Chrome or Microsoft Edge**, recent desktop version. The exhibit uses **Web Serial**; Firefox and Safari are not suitable.
2. **Python 3**, from https://www.python.org/downloads/windows/ . During installation, enable **Add Python to PATH** if offered. Python is used only to serve the local HTML file; no Python packages are required.
3. **Arduino IDE 2**, from https://www.arduino.cc/en/software . Needed for first-time uploading or changing the Arduino sketch; **not needed for normal operation after upload**.
4. In Arduino IDE, install the **Pololu VL53L0X** library: **Tools** → **Manage Libraries…**, search `VL53L0X`, install the library by **Pololu**. `Wire` and `Servo` are built into the Arduino environment for Uno boards.
5. If Windows does not recognise the Arduino as a serial/COM device, install the appropriate USB-serial driver for the specific Uno-compatible board (some clones use CH340/CH341). Do not install random drivers unless Device Manager indicates they are needed.

No account, cloud backend, API key, npm, Node.js, or Python serial package is needed. Internet is useful for installing the software and loading the optional web fonts, but the voting connection itself is local USB.

3. Wiring (POWER OFF while changing wires)

Both VL53L0X sensors share the I²C bus. Their **XSHUT** pins are separate so the sketch can assign different addresses at startup.

· Connection · Arduino / power connection ·

· --- · --- ·

· Hand sensor VIN · Arduino 5 V (only if sensor breakout is 5 V compatible) ·

· Hand sensor GND · Arduino GND ·

· Hand sensor SDA · Arduino A4 ·

· Hand sensor SCL · Arduino A5 ·

· Hand sensor XSHUT · Arduino D2 ·

· Voting sensor VIN · Arduino 5 V (only if breakout is 5 V compatible) ·

· Voting sensor GND · Arduino GND ·

· Voting sensor SDA · Arduino A4 (shared) ·

· Voting sensor SCL · Arduino A5 (shared) ·

· Voting sensor XSHUT · Arduino D3 ·

· Servo orange/yellow (signal) · Arduino D9 ·

· Servo red (+) · **Recommended:** separate regulated 5 V supply (+) ·

· Servo brown/black (–) · Separate 5 V supply GND (–) ·

· Separate supply GND (–) · **Also connect to Arduino GND** (common ground) ·

Safety: Do not connect the external supply's +5 V to the Arduino 5 V pin when the Arduino is USB-powered. Do not apply 5 V to bare 3.3 V-only sensors. The sketch releases XSHUT as an input instead of driving it HIGH at 5 V. Confirm power-supply polarity before connecting. A servo that clicks but does not turn may be underpowered or jammed.

4. First-time Arduino setup (once per board / after code changes)

1. Assemble and inspect the wiring with USB disconnected.
2. Plug the Arduino into the laptop using a **USB data cable**.
3. Open Arduino IDE and open `CommunityCollage_Combined/CommunityCollage_Combined.ino`.
4. Select **Tools** → **Board** → **Arduino Uno** (or the board matching the DFRduino Uno-compatible device).
5. Select **Tools** → **Port** → **COMx**, choosing the port that appears when the Arduino is plugged in.
6. Click **Upload**. Wait for the successful upload message.
7. Open **Serial Monitor**, set **115200 baud**, and look for `READY`. If you see `ERROR:HAND_SENSOR` or `ERROR:VOTE_SENSOR`, check XSHUT, SDA, SCL, power and ground.
8. Check hand-trigger operation and that the servo opens and closes. Then test a ball through each voting hole; expected serial lines are `VOTE:1`, `VOTE:2`, `VOTE:3`, and `VOTE:4`.
9. **Close Serial Monitor and Serial Plotter** before connecting from the browser: only one program can normally own the COM port at a time.

Important: The bundled combined sketch is based on the last combined code discussed, including the calibrated voting distances. It is provided for deployment, but **its servo operation has not been validated with the final power arrangement**. The earlier `arduino_ball_votes.ino` was voting-only; do not use that file if you need hand-triggered ball release.

5. Everyday startup — operator quick start

1. Connect the assembled Arduino to the laptop by USB. Connect the servo's external regulated 5 V supply if used. Confirm the exhibit is mechanically clear.
2. Open the `Community-Collage` folder in Windows File Explorer.
3. Click the File Explorer address bar, type `cmd`, then press **Enter**. This opens Command Prompt in the correct folder.
4. Run:

```bat
py -m http.server 8000
```

If `py` is not recognised, try `python -m http.server 8000`.

5. **Keep the Command Prompt window open.** It should say it is serving HTTP on port 8000.
6. Open **Chrome or Edge** and navigate to:

`http://localhost:8000/community-collage-arduino-no-popups.html`

7. Click **Connect Arduino**, select the correct Arduino COM port, and approve the browser prompt. Wait for the connected status.
8. Test one ball. Confirm one ball appears on the screen and the response counter increases by one.
9. For presentation mode, press **F11** to make the browser full-screen. Keep the browser tab, USB cable and Command Prompt open throughout the exhibit.
10. To start with a fresh count, **refresh the page before visitors begin**. Refreshing clears the current results. Avoid refreshing during the event.

Operator note: On each new computer/browser, a person must initially click **Connect Arduino** and grant permission. It is not automatically connected simply because USB is plugged in. The Arduino may reset briefly when the browser opens the serial port; this is expected.

6. What the exhibit does

- Hand sensor: detects a hand nearer than **150 mm** and commands the servo to open; it rearms after the hand moves beyond **200 mm**.
- Servo: configured to move between **0° closed** and **90° open**, staying open for **300 ms** in the bundled sketch. These settings may need physical calibration for the actual mechanism.
- Voting sensor: identifies four calibrated ball distances in millimetres:

- Vote 1: **180–210 mm**
- Vote 2: **240–270 mm**
- Vote 3: **300–320 mm**
- Vote 4: **360–380 mm**

- Arduino sends `VOTE:1` to `VOTE:4` at **115200 baud**.
- Browser reads those messages and calls the same animation/vote logic used by keyboard keys **1–4**.
- Duplicate prevention: after a vote, the sensor must read an empty path **above 500 mm** for **150 ms**, and at least **600 ms** must have passed since the vote. **The 500 mm empty-path threshold is an assumption that may require adjustment.**
- The latest website does **not** show the intermediate text pop-ups. It does show the final results screen when the configured target is reached (default **75 responses**).
- Key **R** restarts after the final screen; refreshing the browser resets the session at any time.

7. Troubleshooting

· Problem · Check / fix ·

· --- · --- ·

· `localhost refused to connect` · Start `py -m http.server 8000`; leave Command Prompt open. ·

· `404 Not Found` · Confirm the HTML filename matches the URL and Command Prompt was opened in its folder. ·

· `py` not recognised · Try `python -m http.server 8000`; otherwise install Python 3. ·

· Port 8000 already in use · Run `py -m http.server 8001`, then visit `http://localhost:8001/...`. ·

· No **Connect Arduino** option / unsupported browser · Use current desktop Chrome or Edge and `http://localhost` (not a `file:///` page). ·

· Arduino COM port missing · Check USB **data** cable, USB port, Device Manager, and the board's USB-serial driver. ·

· Browser says port busy / cannot open · Close Arduino IDE Serial Monitor/Plotter and other apps using the COM port. Disconnect/reconnect if needed. ·

· Browser connected but no votes · Test Serial Monitor separately at 115200 for `VOTE:n`; then close it and reconnect in the browser. Check ball sensor alignment. ·

· Only the first ball counts · Empty-path reading may not exceed 500 mm; adjust `CLEAR_DISTANCE` after measuring the empty path, then upload again. ·

· Wrong voting option · Recheck sensor position and the four calibrated distance bands. ·

· Hand distance responds but servo only clicks · Known unresolved issue: servo works alone but not with everything powered together. Check for mechanical binding and use a separate regulated 5 V supply with common ground. A 4.7 V multimeter reading alone does not prove adequate peak current. ·

· Servo moves opposite way / not far enough · Adjust `CLOSED_ANGLE`, `OPEN_ANGLE`, and possibly `OPEN_TIME` in the sketch; reupload. ·

· `ERROR:HAND_SENSOR` / `ERROR:VOTE_SENSOR` · Check both sensors' power, shared A4/A5 bus, and separate D2/D3 XSHUT wiring. ·

· QR image missing · Copy `https_emaie3_github_io_2850_ (1).png` into the same folder as the HTML. ·

· Fonts look different offline · Google Fonts require internet; the rest of the exhibit can run locally. ·

8. Shutdown

1. End the session; if results are needed, **record them before closing**.
2. Close the browser tab.
3. Stop the Python server with **Ctrl+C** in Command Prompt.
4. Disconnect external servo power, then unplug the Arduino USB cable.

9. Important limitations and handover notes

- **Votes are held in browser memory, not saved to a database or file.** Refreshing, closing the tab, or losing power loses the current session. Record the results before shutting down.
- A laptop must stay awake and connected; disable automatic sleep and connect its charger.
- The browser must stay connected to the Arduino's serial port.
- The QR image is a separate asset and is **not** embedded in the HTML.
- Arduino IDE is required only for uploading/debugging, not for routine event operation.
- **Before the event:** confirm the servo works under full load, test at least two consecutive balls, and check all four voting options.
- If the servo remains unreliable, do not depend on automatic ball release. The voting display can still operate with manually released balls, provided the voting sensor is working.